Practice Problems: Instrumental Variables

Quantitative Economics

Swapnil Singh

Lietuvos Bankas

Kaunas University of Technology

Vilnius University, 2026

20 problems from simple to tough. Solutions in the appendix.

Overview

20 problems, ordered by difficulty:

- Problems 1–5: Definitions & conceptual foundations
- Problems 6–10: Bias formulas & derivations
- Problems 11–15: TSLS mechanics & consistency
- Problems 16–20: Advanced topics & applications

Each problem slide has a Solution button linking to the appendix.
Each solution slide has a Back button to return.

Problem 1: Three sources of endogeneity

List the three classical sources of endogeneity. For each, give a one-sentence explanation of *why* it causes $\text{cov}(X_i, u_i) \neq 0$.

Solution

Problem 2: Instrument validity conditions

State the two conditions required for a variable Z to be a valid instrument for the endogenous regressor X in the model $Y_i = \beta_0 + \beta_1 X_i + u_i$.
For each condition, explain in one sentence what goes wrong if it fails.

Solution

Problem 3: Endogenous or exogenous?

For each scenario, state whether X is endogenous or exogenous and identify which source of endogeneity (if any) applies:

(a) Regressing wages on years of education, omitting innate ability.

(b) Regressing crop yield on fertilizer use, where fertilizer is randomly assigned across plots.

(c) Regressing quantity demanded on price in a competitive market.

(d) Regressing test scores on hours studied, where hours studied is measured with error.

Solution

Problem 4: Levels of exogeneity

State the three progressively stronger conditions for instrument exogeneity:

(a) Zero correlation

(b) Mean independence

(c) Full independence

Show that mean independence implies zero correlation. Then provide a counterexample demonstrating that the converse is false.

Solution

Problem 5: Evaluating proposed instruments

A researcher wants to estimate the effect of class size on student test scores. She proposes the following instruments. For each, discuss whether relevance and exogeneity are likely satisfied:

(a) School district budget per pupil.

(b) Random fluctuations in cohort size due to birth-date variation.

(c) Average temperature in the school district.

Solution

Problem 6: Omitted variable bias formula

Suppose the true DGP is:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \varepsilon_i, \qquad E(\varepsilon_i \mid X_i, W_i) = 0$$

but W_i is unobserved. Define $u_i = \beta_2 W_i + \varepsilon_i$.

(a) Derive $\text{cov}(X_i, u_i)$ and state conditions under which it is nonzero.

(b) Show that $\widehat{\beta}_1^{OLS} \xrightarrow{p} \beta_1 + \beta_2 \dfrac{\text{cov}(X_i, W_i)}{\text{var}(X_i)}$.

(c) If $\beta_2 > 0$ and $\text{cov}(X, W) > 0$, is the OLS estimate biased upward or downward?

Solution

Problem 7: Attenuation bias

True model: $Y_i = \beta_0 + \beta_1 X_i^* + \varepsilon_i$ with $E(\varepsilon_i \mid X_i^*) = 0$.
We observe $X_i = X_i^* + w_i$ (classical measurement error).

(a) Substitute $X_i^* = X_i - w_i$ and derive the composite error u_i.

(b) Show that $\text{cov}(X_i, u_i) = -\beta_1 \sigma_w^2$.

(c) Derive the probability limit of $\widehat{\beta}_1^{OLS}$ and explain why the bias is called "attenuation bias."

(d) If the true $\beta_1 = 2$, $\sigma_{X^*}^2 = 4$, and $\sigma_w^2 = 1$, compute the plim of OLS.

Solution

Problem 8: Simultaneous equations bias

Consider the supply–demand system:

$$\begin{aligned} \text{Demand}: \quad & Q_i = \alpha_0 + \alpha_1 P_i + u_i^d \qquad (\alpha_1 < 0) \\ \text{Supply}: \quad & Q_i = \gamma_0 + \gamma_1 P_i + u_i^s \qquad (\gamma_1 > 0) \end{aligned}$$

Assume $u^d \perp u^s$, both mean-zero with variances σ_d^2 and σ_s^2.

(a) Solve for the equilibrium price P_i.

(b) Show that $\text{cov}(P_i, u_i^d) = \dfrac{-\sigma_d^2}{\alpha_1 - \gamma_1}$ and explain its sign.

(c) Derive: $\widehat{\alpha}_1^{OLS} \xrightarrow{p} \dfrac{\alpha_1 \sigma_s^2 + \gamma_1 \sigma_d^2}{\sigma_s^2 + \sigma_d^2}$.

(d) Interpret this result: what does OLS estimate?

Solution

Problem 9: Exogeneity failure and TSLS inconsistency

Suppose the instrument Z is not perfectly exogenous: $\text{cov}(Z_i, u_i) = \rho \neq 0$.

(a) Show that $\widehat{\beta}_1^{TSLS} \xrightarrow{p} \beta_1 + \dfrac{\rho}{\text{cov}(Z_i, X_i)}$.

(b) If $\rho = 0.01$ and $\text{cov}(Z_i, X_i) = 0.5$, what is the asymptotic bias?

(c) Now suppose the instrument is weak: $\text{cov}(Z_i, X_i) = 0.02$. Recompute the asymptotic bias. What do you observe?

(d) Explain in words why weak instruments *amplify* violations of exogeneity.

Solution

Problem 10: Direction of OVB with IV

Return to the OVB setup. The true DGP is $Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \varepsilon_i$. An instrument Z satisfies $\text{cov}(Z, X) \neq 0$ and $\text{cov}(Z, \varepsilon) = 0$.

(a) Write the exogeneity condition $\text{cov}(Z_i, u_i) = 0$ in terms of β_2 and $\text{cov}(Z_i, W_i)$.

(b) Why does this mean the instrument must be uncorrelated with the omitted variable W?

(c) A researcher proposes parental income as an instrument for education in a wage regression (omitting ability). Is this a good instrument? Why or why not?

Solution

Problem 11: The TSLS formula

In the simple model $Y_i = \beta_0 + \beta_1 X_i + u_i$ with one instrument Z:

(a) The TSLS estimator is the OLS estimator of Y on $\widehat{X}$, where $\widehat{X}_i = \widehat{\pi}_0 + \widehat{\pi}_1 Z_i$. Starting from $\widehat{\beta}_1^{TSLS} = \frac{s_{\widehat{X}Y}}{s_{\widehat{X}}^2}$, show that $\widehat{\beta}_1^{TSLS} = \frac{s_{ZY}}{s_{ZX}}$.

(b) Explain why this is called the "Wald estimator" and interpret it as a ratio of a reduced-form effect to a first-stage effect.

Solution

Problem 12: Consistency of TSLS

Starting from $Y_i = \beta_0 + \beta_1 X_i + u_i$:

(a) Take population covariances of both sides with Z_i and solve for β_1 under the IV assumptions.

(b) Use the law of large numbers and continuous mapping theorem to show that $\widehat{\beta}_1^{TSLS} = s_{ZY}/s_{ZX} \xrightarrow{p} \beta_1$.

(c) At which step does instrument relevance play a role? At which step does instrument exogeneity play a role?

Solution

Problem 13: First-stage F-statistic

A researcher estimates a first-stage regression and obtains a first-stage F-statistic of 7.2.

(a) Is the instrument weak according to the rule of thumb?

(b) Using the approximate bias formula $E(\widehat{\beta}_1^{TSLS}) - \beta_1 \approx \dfrac{\beta_1^{OLS} - \beta_1}{E(F) - 1}$, what fraction of the OLS bias does the TSLS estimate retain?

(c) List three strategies the researcher could pursue.

(d) Why are standard TSLS confidence intervals unreliable with weak instruments?

Solution

Problem 14: Identification and the order condition

Consider the general IV model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 W_{1i} + u_i$$

where X_1, X_2 are endogenous and W_1 is exogenous.

(a) How many instruments are needed at minimum?

(b) A researcher proposes three instruments Z_1, Z_2, Z_3. Is the model exactly identified, overidentified, or underidentified?

(c) Describe what the first-stage regressions look like. How many are there?

(d) Can the researcher test instrument exogeneity? If so, what is the degrees of freedom of the J-test?

Solution

Problem 15: The J-test

A researcher estimates a model with $k = 1$ endogenous regressor and $m = 3$ instruments. After estimating by TSLS:

(a) Describe the steps to compute the J-statistic.

(b) Under H_0, what is the distribution of the J-statistic? State the degrees of freedom.

(c) The researcher obtains $J = 7.8$. At the 5% significance level, does she reject? (Use $\chi^2_2(0.05) = 5.99$.)

(d) The J-test rejects. Does this mean *all* instruments are endogenous? Explain.

Solution

Problem 16: Sampling distribution of TSLS

From the representation $\widehat{\beta}_1^{TSLS} = \beta_1 + \dfrac{\frac{1}{n}\sum(Z_i - \bar{Z})u_i}{\frac{1}{n}\sum(Z_i - \bar{Z})(X_i - \bar{X})}$:

(a) Define $\bar{q} = \frac{1}{n}\sum(Z_i - \bar{Z})u_i$. Show that $E(\bar{q}) = 0$ under instrument exogeneity and state which theorem implies $\sqrt{n}\bar{q} \xrightarrow{d} N(0, \sigma_q^2)$.

(b) The denominator converges to $\text{cov}(Z, X)$ in probability. Which theorem combines numerator and denominator to yield asymptotic normality of $\widehat{\beta}_1^{TSLS}$?

(c) Write down the asymptotic variance $\sigma^2_{\widehat{\beta}_1^{TSLS}}$ and explain why it is larger when $\text{cov}(Z, X)$ is small.

Solution

Problem 17: Cigarette demand application

Using panel data (10-year differences), a researcher estimates:

$$\Delta \ln Q_i = \beta_0 + \beta_1 \Delta \ln P_i + \beta_2 \Delta \ln \text{Inc}_i + u_i$$

with instruments ΔSalesTax and ΔCigTax.
Results: $\widehat{\beta}_1^{(1)} = -0.94$ (SalesTax only), $\widehat{\beta}_1^{(2)} = -1.34$ (CigTax only), $\widehat{\beta}_1^{(3)} = -1.20$ (both). The J-test p-value with both instruments is 0.026.

(a) First-stage F-statistics are 33.7, 107.2, and 88.6. Are instruments weak?

(b) Interpret the J-test result. What does rejection imply?

(c) Which instrument would you prefer and why? What is the economic argument?

(d) Interpret the preferred elasticity estimate in terms of policy.

Solution

Problem 18: Anderson–Rubin test

The Anderson–Rubin (AR) test of $H_0 : \beta_1 = \beta_{1,0}$ works by:

1. Constructing $Y_i^* = Y_i - \beta_{1,0} X_i$
2. Regressing Y_i^* on Z's and W's
3. Testing whether all coefficients on Z's are jointly zero

(a) Explain intuitively why, if $\beta_1 = \beta_{1,0}$ and the instruments are exogenous, the Z's should have no predictive power for Y_i^*.

(b) Why is this test valid even when instruments are weak?

(c) Describe how to construct a confidence set by inverting the AR test.

(d) What unusual features can AR confidence sets have?

Solution

Problem 19: IV with simultaneous causality – full derivation

In the supply–demand system with a supply shifter Z:

$$\begin{aligned} \text{Demand}: \quad & Q_i = \alpha_0 + \alpha_1 P_i + u_i^d \\ \text{Supply}: \quad & Q_i = \gamma_0 + \gamma_1 P_i + \delta Z_i + u_i^s \end{aligned}$$

Assume $\text{cov}(Z_i, u_i^d) = 0$, $\text{cov}(Z_i, u_i^s) = 0$, and $\delta \neq 0$.

(a) Solve for the equilibrium price P_i in terms of Z_i and the shocks.

(b) Verify that $\text{cov}(Z_i, P_i) \neq 0$ (relevance).

(c) Verify that $\text{cov}(Z_i, u_i^d) = 0$ (exogeneity for the demand equation).

(d) Compute $\text{cov}(Z_i, Q_i)$ using the demand equation and show that $\dfrac{\text{cov}(Z_i, Q_i)}{\text{cov}(Z_i, P_i)} = \alpha_1$, confirming TSLS identifies the demand slope.

Solution

Problem 20: Comprehensive analysis

A health economist estimates the effect of hospital spending (X) on patient survival (Y). She suspects spending is endogenous because sicker patients receive more spending.
She proposes two instruments: (i) distance to nearest teaching hospital (Z_1), (ii) local property tax revenue (Z_2). Her results:

- OLS: $\widehat{\beta}_1^{OLS} = -0.05$ (SE $= 0.01$)
- TSLS (both IVs): $\widehat{\beta}_1^{TSLS} = 0.12$ (SE $= 0.04$)
- First-stage $F = 14.3$
- J-statistic $= 0.45$, p-value $= 0.50$

(a) Why is OLS likely biased? In which direction?

(b) Assess instrument strength. Are the instruments weak?

(c) Interpret the J-test. Does it validate both instruments?

(d) Discuss the exogeneity of each instrument. Which is more credible?

(e) Reconcile the OLS and TSLS estimates economically.

Solution

Solutions

Solution 1: Three sources of endogeneity

1. **Omitted Variable Bias**: An unobserved variable W in the error u is correlated with X, so $\text{cov}(X, u) = \beta_2 \,\text{cov}(X, W) \neq 0$.
2. **Measurement Error**: The observed $X = X^* + w$ contains noise w that mechanically enters both X and $u = \varepsilon - \beta_1 w$, creating $\text{cov}(X, u) = -\beta_1 \sigma_w^2 \neq 0$.
3. **Simultaneous Causality**: X and Y are jointly determined in equilibrium, so the structural error u feeds back into X through the equilibrium mapping.

Back to Problem 1

Solution 2: Instrument validity conditions

Condition 1: Instrument Relevance

$\text{corr}(Z_i, X_i) \neq 0$. Z must be correlated with X.
If it fails: Z carries no information about X. The denominator $\text{cov}(Z, X) \to 0$, making the TSLS estimator undefined (ratio of two normals, not consistent).

Condition 2: Instrument Exogeneity

$\text{corr}(Z_i, u_i) = 0$. Z must be uncorrelated with the error.
If it fails: TSLS converges to $\beta_1 + \text{cov}(Z, u)/\text{cov}(Z, X) \neq \beta_1$. The estimator is inconsistent — it "locks onto" contaminated variation.

Back to Problem 2

Solution 3: Endogenous or exogenous?

(a) **Endogenous** — *Omitted variable bias*. Ability is in u, and $\text{cov}(\text{education}, \text{ability}) > 0$.

(b) **Exogenous**. Random assignment ensures $\text{cov}(X, u) = 0$. OLS is consistent.

(c) **Endogenous** — *Simultaneous causality*. Price and quantity are jointly determined by supply and demand; OLS on either equation is inconsistent.

(d) **Endogenous** — *Measurement error*. Observed $X = X^* + w$, and the noise w enters both X and the composite error, creating $\text{cov}(X, u) = -\beta_1 \sigma_w^2 \neq 0$.

Back to Problem 3

Solution 4: Levels of exogeneity

Three levels: (a) $\text{cov}(Z, u) = 0$; (b) $E(u \mid Z) = 0$; (c) $u \perp\!\!\!\perp Z$.
Proof that (b) $\Rightarrow$ (a):

$$E(Zu) = E[E(Zu \mid Z)] = E[Z \cdot \underbrace{E(u \mid Z)}_{=0}] = 0$$

$$E(u) = E[\underbrace{E(u \mid Z)}_{=0}] = 0$$

So $\text{cov}(Z, u) = E(Zu) - E(Z)E(u) = 0$.
Counterexample (converse fails): Let $Z \sim N(0, 1)$, $u = Z^2 - 1$.

- $E(u) = 0$, $\text{cov}(Z, u) = E(Z^3) = 0$. ✓
- But $E(u \mid Z = z) = z^2 - 1 \neq 0$ for $z \neq \pm 1$. ✗

Back to Problem 4

Solution 5: Evaluating proposed instruments

(a) **Budget per pupil**: *Relevant* (more funding $\rightarrow$ smaller classes), but likely *not exogenous* — budget correlates with district wealth, parental involvement, and school quality, all of which affect test scores directly.

(b) **Random birth-date fluctuations** (Hoxby, 2000): *Relevant* (more students $\rightarrow$ larger classes) and plausibly *exogenous* — biological variation in birth dates is unrelated to school quality or family background. Good instrument.

(c) **Average temperature**: Likely *irrelevant* (no clear mechanism linking temperature to class size). Even if correlated, the link is too weak. Not a useful instrument.

Back to Problem 5

Solution 6: Omitted variable bias formula

(a) $\text{cov}(X, u) = \text{cov}(X, \beta_2 W + \varepsilon) = \beta_2 \,\text{cov}(X, W) + \underbrace{\text{cov}(X, \varepsilon)}_{=0} =$

$\beta_2 \,\text{cov}(X, W)$.

Nonzero iff $\beta_2 \neq 0$ *and* $\text{cov}(X, W) \neq 0$.

(b) $\widehat{\beta}_1^{OLS} \xrightarrow{p} \beta_1 + \dfrac{\text{cov}(X, u)}{\text{var}(X)} = \beta_1 + \beta_2 \dfrac{\text{cov}(X, W)}{\text{var}(X)}$. ✓

(c) $\beta_2 > 0$ and $\text{cov}(X, W) > 0 \Rightarrow \text{bias} = \beta_2 \dfrac{\text{cov}(X, W)}{\text{var}(X)} > 0$.

OLS is biased upward — it overestimates β_1.

Back to Problem 6

Solution 7: Attenuation bias

(a) $Y_i = \beta_0 + \beta_1(X_i - w_i) + \varepsilon_i = \beta_0 + \beta_1 X_i + \underbrace{(\varepsilon_i - \beta_1 w_i)}_{u_i}$.

(b) $\text{cov}(X, u) = \text{cov}(X^* + w,\ \varepsilon - \beta_1 w) = -\beta_1 \sigma_w^2$.

(c) $\widehat{\beta}_1^{OLS} \xrightarrow{p} \beta_1 + \dfrac{-\beta_1 \sigma_w^2}{\sigma_{X^*}^2 + \sigma_w^2} = \beta_1 \underbrace{\left(\dfrac{\sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_w^2} \right)}_{<1}$.

The coefficient is "shrunk" toward zero — attenuation bias. More noise ($\sigma_w^2 \uparrow$) $\Rightarrow$ more attenuation.

(d) $\text{plim} = 2 \times \dfrac{4}{4+1} = 2 \times \dfrac{4}{5} = \boxed{1.6}$. OLS underestimates by 20%.

Back to Problem 7

Solution 8: Simultaneous equations bias (1/2)

(a) Set demand = supply: $\alpha_0 + \alpha_1 P + u^d = \gamma_0 + \gamma_1 P + u^s$.
Solve: $P_i = \dfrac{(\gamma_0 - \alpha_0) + (u_i^s - u_i^d)}{\alpha_1 - \gamma_1}$.

(b) $\text{cov}(P, u^d) = \dfrac{1}{\alpha_1 - \gamma_1}[\text{cov}(u^s, u^d) - \text{var}(u^d)] = \dfrac{-\sigma_d^2}{\alpha_1 - \gamma_1}$.
Since $\alpha_1 < 0 < \gamma_1$: $\alpha_1 - \gamma_1 < 0$, so $\text{cov}(P, u^d) > 0$.
Intuition: A positive demand shock raises the equilibrium price.

Continued

Back to Problem 8

Solution 8: Simultaneous equations bias (2/2)

(c) $\text{var}(P) = \dfrac{\sigma_s^2 + \sigma_d^2}{(\alpha_1 - \gamma_1)^2}$.

$$\widehat{\alpha}_1^{OLS} \xrightarrow{p} \alpha_1 + \frac{\text{cov}(P, u^d)}{\text{var}(P)} = \alpha_1 + \frac{-\sigma_d^2(\alpha_1 - \gamma_1)}{(\sigma_s^2 + \sigma_d^2)} = \frac{\alpha_1 \sigma_s^2 + \gamma_1 \sigma_d^2}{\sigma_s^2 + \sigma_d^2}$$

(d) This is a variance-weighted average of the demand slope α_1 and the supply slope γ_1. OLS estimates *neither* the demand nor the supply curve — it produces a meaningless hybrid. Only when $\sigma_d^2 = 0$ (no demand shocks) does OLS recover α_1.

Back to Problem 8

Solution 9: Exogeneity failure and TSLS inconsistency

(a) From $\text{cov}(Z, Y) = \beta_1 \text{cov}(Z, X) + \text{cov}(Z, u)$:

$$\frac{\text{cov}(Z, Y)}{\text{cov}(Z, X)} = \beta_1 + \frac{\rho}{\text{cov}(Z, X)}$$

Since $s_{ZY}/s_{ZX} \xrightarrow{p}$ this ratio: $\widehat{\beta}_1^{TSLS} \xrightarrow{p} \beta_1 + \dfrac{\rho}{\text{cov}(Z, X)}$. ✓

(b) Bias $= 0.01/0.5 = \boxed{0.02}$. Small.

(c) Bias $= 0.01/0.02 = \boxed{0.50}$. Twenty-five times larger!

(d) The asymptotic bias is $\rho/\text{cov}(Z, X)$. When the instrument is weak ($\text{cov}(Z, X) \approx 0$), the denominator shrinks, amplifying even tiny exogeneity violations into large biases.

Back to Problem 9

Solution 10: Direction of OVB with IV

(a) $\text{cov}(Z, u) = \beta_2 \,\text{cov}(Z, W) + \text{cov}(Z, \varepsilon) = 0$. Since $\text{cov}(Z, \varepsilon) = 0$ by assumption, we need $\beta_2 \,\text{cov}(Z, W) = 0$.

(b) Since $\beta_2 \neq 0$ (otherwise there is no OVB), we must have $\text{cov}(Z, W) = 0$. The instrument must be uncorrelated with the omitted variable.

(c) **Parental income is a bad instrument.** It is correlated with ability (W): children from wealthier families tend to have advantages (better nutrition, tutoring, etc.) that correlate with ability. So $\text{cov}(Z, W) \neq 0$, violating exogeneity. A better instrument: quarter of birth (Angrist & Krueger, 1991).

Back to Problem 10

Solution 11: The TSLS formula

(a) Since $\widehat{X}_i = \widehat{\pi}_0 + \widehat{\pi}_1 Z_i$, deviations are $\widehat{X}_i - \bar{\widehat{X}} = \widehat{\pi}_1(Z_i - \bar{Z})$. Substitute:

$$s_{\widehat{X}Y} = \frac{1}{n-1}\sum \widehat{\pi}_1(Z_i - \bar{Z})(Y_i - \bar{Y}) = \widehat{\pi}_1\, s_{ZY}$$

$$s^2_{\widehat{X}} = \frac{1}{n-1}\sum \widehat{\pi}_1^2(Z_i - \bar{Z})^2 = \widehat{\pi}_1^2\, s_Z^2$$

$$\widehat{\beta}_1^{TSLS} = \frac{\widehat{\pi}_1 s_{ZY}}{\widehat{\pi}_1^2 s_Z^2} = \frac{s_{ZY}}{\widehat{\pi}_1 s_Z^2} = \frac{s_{ZY}}{s_{ZX}}, \quad \text{since } \widehat{\pi}_1 = s_{ZX}/s_Z^2. \checkmark$$

(b) **Wald estimator**: s_{ZY} is the *reduced-form* effect ($Z \to Y$) and s_{ZX} is the *first-stage* effect ($Z \to X$). The ratio recovers the structural effect $X \to Y$: "how much Y moves per unit of X movement caused by Z."

Back to Problem 11

Solution 12: Consistency of TSLS

(a) $\text{cov}(Z, Y) = \text{cov}(Z, \beta_0 + \beta_1 X + u) = \beta_1 \,\text{cov}(Z, X) + \underbrace{\text{cov}(Z, u)}_{=0 \text{ (exogeneity)}}$.

Solve: $\beta_1 = \dfrac{\text{cov}(Z, Y)}{\text{cov}(Z, X)}$ (requires $\text{cov}(Z, X) \neq 0$: relevance).

(b) By LLN: $s_{ZY} \xrightarrow{p} \text{cov}(Z, Y)$, $s_{ZX} \xrightarrow{p} \text{cov}(Z, X)$.

By continuous mapping theorem (Slutsky): $\dfrac{s_{ZY}}{s_{ZX}} \xrightarrow{p} \dfrac{\text{cov}(Z, Y)}{\text{cov}(Z, X)} = \beta_1$. ✓

(c) **Exogeneity**: used in (a) to set $\text{cov}(Z, u) = 0$, allowing isolation of β_1.
Relevance: used in (a) to divide by $\text{cov}(Z, X) \neq 0$, and in (b) to apply Slutsky (denominator converges to a nonzero constant, not zero).

Back to Problem 12

Solution 13: First-stage F-statistic

(a) Yes, $F = 7.2 < 10$. The instrument is weak by the Stock–Yogo rule of thumb.

(b) Fraction of OLS bias $\approx \dfrac{1}{E(F)-1} = \dfrac{1}{7.2-1} = \dfrac{1}{6.2} \approx \boxed{16.1\%}$. TSLS retains about 16% of the OLS bias — substantial.

(c) Three strategies:

- Find stronger instruments (requires domain knowledge).
- If overidentified, discard the weakest instruments.
- Use weak-instrument-robust inference: Anderson–Rubin or CLR tests/confidence sets.

(d) With weak instruments, the TSLS distribution is not well-approximated by the normal. It is heavy-tailed and centered closer to the OLS plim, so standard t-tests and CIs have incorrect size (true coverage $\ll$ nominal 95%).

Back to Problem 13

Solution 14: Identification and the order condition

(a) $k = 2$ endogenous regressors $\Rightarrow$ need $m \geq 2$ instruments. **Minimum: 2 instruments.**

(b) $m = 3 > k = 2$: the model is overidentified (one extra instrument).

(c) Two first-stage regressions (one per endogenous variable):

$$X_{1i} = \pi_0 + \pi_1 Z_{1i} + \pi_2 Z_{2i} + \pi_3 Z_{3i} + \pi_4 W_{1i} + v_{1i}$$
$$X_{2i} = \delta_0 + \delta_1 Z_{1i} + \delta_2 Z_{2i} + \delta_3 Z_{3i} + \delta_4 W_{1i} + v_{2i}$$

Each includes all instruments and all exogenous regressors.

(d) Yes, because $m > k$. The J-test has $m - k = 3 - 2 = \boxed{1}$ degree of freedom. Under H_0: $J \sim \chi^2_1$.

Back to Problem 14

Solution 15: The J-test

(a) Steps:

1. Estimate the model by TSLS; obtain residuals $\widehat{u}_i^{TSLS}$.
2. Regress $\widehat{u}_i^{TSLS}$ on all Z's and W's. Compute the F-statistic testing all Z-coefficients $= 0$.
3. $J = m \times F$.

(b) $J \sim \chi^2_{m-k} = \chi^2_{3-1} = \chi^2_2$.

(c) $J = 7.8 > 5.99 = \chi^2_2(0.05)$. Reject H_0 at the 5% level.

(d) No. The J-test rejects the hypothesis that *all* instruments are exogenous, but it could be that only *one* is endogenous. It does not identify *which* instrument(s) are problematic. Moreover, the test is conditional on at least k instruments being valid — it can never confirm the baseline identifying assumption.

Back to Problem 15

Solution 16: Sampling distribution of TSLS

(a) $E(\bar{q}) \approx E\left[\frac{1}{n}\sum(Z_i - \mu_Z)u_i\right] = E[(Z_i - \mu_Z)u_i] = \text{cov}(Z, u) = 0$ under exogeneity.
Since $q_i = (Z_i - \mu_Z)u_i$ are i.i.d. with $E(q_i) = 0$ and $\text{var}(q_i) = \sigma_q^2 < \infty$, the Central Limit Theorem gives $\sqrt{n}\,\bar{q} \xrightarrow{d} N(0, \sigma_q^2)$.

(b) **Slutsky's theorem**: since the denominator $\xrightarrow{p} \text{cov}(Z, X) \neq 0$ (a constant), the ratio of an asymptotically normal numerator and a probability limit in the denominator is asymptotically normal.

(c) $\sigma^2_{\widehat{\beta}_1^{TSLS}} = \dfrac{1}{n} \cdot \dfrac{\text{var}[(Z_i - \mu_Z)u_i]}{[\text{cov}(Z, X)]^2}$.
When $\text{cov}(Z, X)$ is small, the denominator shrinks, inflating the variance. Weak instruments $\Rightarrow$ imprecise estimates.

Back to Problem 16

Solution 17: Cigarette demand application

(a) All F-statistics $\gg 10$ (33.7, 107.2, 88.6). Instruments are not weak.

(b) J-test $p = 0.026 < 0.05$: reject H_0 that both instruments are exogenous. At least one instrument is likely endogenous.

(c) **Prefer SalesTax** ($\widehat{\beta}_1 = -0.94$). General sales tax is set for broad public finance reasons, unrelated to the cigarette market. Cigarette-specific tax may be influenced by tobacco industry lobbying and is thus more likely correlated with demand-side factors.

(d) A 1% price increase reduces cigarette consumption by about 0.94% in the long run. Demand is approximately unit-elastic at the 10-year horizon. To reduce consumption by 20%, prices need to rise by $\approx 20/0.94 \approx 21\%$.

Back to Problem 17

Solution 18: Anderson–Rubin test

(a) If $\beta_1 = \beta_{1,0}$, then $Y_i^* = Y_i - \beta_{1,0} X_i = \beta_0 + u_i + \beta_2 W_{1i} + \cdots$ Under exogeneity, $\text{cov}(Z, u) = 0$. So Z has no predictive power for Y^* beyond W's — the F-statistic should be small.

(b) Standard TSLS inference relies on the denominator converging to a nonzero constant (Slutsky). With weak instruments, this fails. The AR test uses the F-distribution of a regression, which is valid regardless of instrument strength — it never divides by $\text{cov}(Z, X)$.

(c) For each candidate $\beta_{1,0}$, run the AR test. The 95% confidence set is $\{\beta_{1,0} : \text{AR test does not reject at } 5\%\}$.

(d) AR confidence sets can be: *empty* (no $\beta_{1,0}$ is consistent with the data), *unbounded* (instruments are too weak to pin down β_1), or *disconnected* (two disjoint intervals).

Back to Problem 18

Solution 19: IV with simultaneous causality (1/2)

(a) Set demand = supply:

$$\alpha_0 + \alpha_1 P + u^d = \gamma_0 + \gamma_1 P + \delta Z + u^s$$

Solve: $\boxed{P_i = \dfrac{(\gamma_0 - \alpha_0) + \delta Z_i + (u_i^s - u_i^d)}{\alpha_1 - \gamma_1}}$.

(b) $\text{cov}(Z, P) = \dfrac{\delta\,\text{var}(Z) + \text{cov}(Z, u^s) - \text{cov}(Z, u^d)}{\alpha_1 - \gamma_1} = \dfrac{\delta\,\text{var}(Z)}{\alpha_1 - \gamma_1} \neq 0$

since $\delta \neq 0$ and $\text{var}(Z) > 0$. Relevance holds. ✓

(c) $\text{cov}(Z, u^d) = 0$ by assumption. Exogeneity holds. ✓

Continued

Back to Problem 19

Solution 19: IV with simultaneous causality (2/2)

(d) From the demand equation: $Q_i = \alpha_0 + \alpha_1 P_i + u_i^d$.
Take covariances with Z:

$$\text{cov}(Z, Q) = \alpha_1 \,\text{cov}(Z, P) + \underbrace{\text{cov}(Z, u^d)}_{=0}$$

Therefore:

$$\frac{\text{cov}(Z, Q)}{\text{cov}(Z, P)} = \alpha_1 \qquad \checkmark$$

TSLS consistently estimates the demand slope α_1 by using only the supply-driven variation in prices (caused by Z). The demand shocks u^d are "projected out."

Back to Problem 19

Solution 20: Comprehensive analysis (1/2)

(a) OLS is biased downward. Sicker patients receive more spending ($\text{cov}(X, u) < 0$ because sicker patients have worse outcomes, i.e., u captures unobserved health). This is a combination of OVB (unobserved severity) and simultaneous causality (spending responds to health status). OLS shows spending *hurts* (-0.05), but this reflects selection, not causation.

(b) $F = 14.3 > 10$. Instruments pass the weak-instrument test. TSLS standard inference is reliable.

(c) $J = 0.45$, $p = 0.50$: do not reject H_0 that both instruments are exogenous. This is *consistent with* validity but does not *prove* it — the J-test can only detect inconsistencies between instruments, not absolute violations.

Continued

Back to Problem 20

Solution 20: Comprehensive analysis (2/2)

(d) **Distance to teaching hospital** (Z_1): Plausibly exogenous — residential location is largely determined before a heart attack occurs. Relevance: patients closer to teaching hospitals receive more intensive (expensive) care. Concern: residential sorting (healthier/wealthier people may live near hospitals).
Property tax revenue (Z_2): Relevant (higher tax revenue $\rightarrow$ better-funded hospitals $\rightarrow$ more spending). But potentially endogenous: wealthy areas have higher property taxes *and* healthier populations. Z_1 is more credible.

(e) OLS: $\widehat{\beta}_1 = -0.05$ (spending appears harmful — selection bias dominates). TSLS: $\widehat{\beta}_1 = 0.12$ (spending improves survival). The sign reversal is exactly what we expect when the dominant endogeneity is negative (sicker patients get more spending). After removing this contamination, IV reveals the true positive causal effect of hospital spending on survival.

Back to Problem 20